# Lingxu Liu


**Applied AI and Agent Engineer | Backend Engineer**

Shenzhen, China | 13011068922 | 15344555700@163.com | github.com/RiderVVV | ridervvv.github.io/resume


## PROFESSIONAL SUMMARY

Backend engineer with 8+ years of experience building scalable systems across AI marketing, IoT, and education technology. Strong background in Golang, distributed services, real-time data processing, and system optimization. Currently focusing on applied AI, agent workflow orchestration, and practical AI product development through postgraduate study in Data Science and Machine Learning for Business.

## SKILLS

**Languages** Python, Golang, SQL

**AI / Data** Machine Learning, Deep Learning, NLP, Text Classification, Image Classification, Data Preprocessing, Model Evaluation

**Engineering** Distributed Systems, RESTful APIs, ETL Pipelines, Performance Optimization, Rule Engines, CI/CD

**Tools** PyTorch, pandas, scikit-learn, Spark, Gradio, Git, Jupyter Notebook, TF-IDF, Logistic Regression, ResNet18

## EDUCATION

**MSc Student in Data Science | Lingnan University** Hong Kong | Sep 2025 - Present

Focusing on machine learning, deep learning, big data analytics, and practical AI product development.

- Built and presented coursework projects in document classification, image classification, and applied AI workflows.
- Strengthened applied skills in NLP, model evaluation, data pipelines, and AI system design through hands-on assignments and group project leadership.
- Combined prior backend engineering experience with postgraduate study in Machine Learning for Business to transition toward applied AI and agent engineering roles.

## PROFESSIONAL EXPERIENCE

**Backend Engineer | Shenzhen Donson Times Information Technology** Shenzhen | Jul 2024 - Aug 2025

Worked on AI marketing products including inSai Hilight, an AI-native marketing video multi-agent platform.

- Supported three major product iterations and contributed to key feature delivery, cross-team implementation, and technical solution design for release cycles.
- Refactored content collection modules, reducing code complexity by 50% and lowering coupling between KOL account and homepage link logic by 40%.
- Improved maintainability and delivery efficiency, shortening new feature development cycles by 30% after refactoring.
- Contributed to engineering efficiency and data infrastructure improvements, including hot-reload workflow adoption and Gauss-to-Doris migration work for large content and brand tables.

**Golang Developer | Wanjiaan IoT Technology** Shenzhen | May 2021 - Apr 2024

Worked on AIoT cloud platform development within the SaaS team.

- Designed and implemented scalable backend services for IoT device access, telemetry collection, and real-time data processing.

- Built ETL pipelines and optimized concurrency, caching, and distributed processing to improve throughput and platform stability.
- Developed rule-engine, automation, alerting, remote management, and firmware upgrade capabilities for SaaS platform operations.
- Improved security through authentication, authorization, and data transmission enhancements.
- Led the IoTHub access platform and Alert and Notification System, supporting 10 million+ device connections and close to 1 million computation tasks per second.

### Golang Developer | Beijing Limi Technology

Beijing | Sep 2016 - Mar 2021

Worked on education platform backend systems, including order, product, user, and monitoring services.

- Designed and developed high-performance order services covering creation, payment, cancellation, and refund flows.
- Integrated third-party payment systems and implemented distributed transaction handling for data consistency.
- Built user management, course management, and monitoring features for online education operations.
- Improved scalability with microservices, containerization, database tuning, caching, and message queues.

## SELECTED PROJECTS

### Document Classification Big Data Pipeline

2026

- Developed a document classification pipeline for invoices, inventory reports, purchase orders, and shipping orders using TF-IDF and Logistic Regression.
- Achieved 99.81% test accuracy and 0.9963 macro F1 while serving as team leader for modeling, integration, and final presentation.

### Waste Classifier with ResNet18

2026

- Built an image classification system for organic vs. recyclable waste using PyTorch and transfer learning.
- The final ResNet18 model achieved 92.88% accuracy and 0.9144 F1, and was deployed through a Gradio demo interface.

## EDUCATION

**Lingnan University** — Expected Jun 2026

MSc in Data Science

**Harbin University** — Jun 2016

Bachelor's in Accounting